

TravelBuddy Chatbot – Coursework Report

1. Chatbot Title:
TravelBuddy: Travel Assistant Chatbot

2. Chatbot Description:

TravelBuddy is a Python based, chatbot designed to help users with travel planning and booking. The chatbot uses a JSON dataset with intents, regex pattern matching and a small knowledge of travel to provide continents, country information, general travel advice, and a personalised response based on the input. It also includes a simple memory mechanism that will remember the user's name and can remember it and recall it later during the conversation when asked. The system uses functions for loading data, pattern matching, generating responses, and running the main loop.

3. Conversation Script (14+ lines)

Transcript from the chatbot run

```
TravelBuddy: Hello! I'm TravelBuddy – your travel assistant.
You: Hi
TravelBuddy: Hello traveler! Where would you like to go next?

You: I want to go to Portugal
TravelBuddy: I don't have detailed information about Portugal yet, but I can still help you plan! What would you like to know – flights, weather, budget, or things to do?

You: How are you?
TravelBuddy: I'm doing great, thank you! How about you?

You: What is your name?
TravelBuddy: I'm TravelBuddy, your personal trip planner.

You: My name is bibi
TravelBuddy: Hello bibi! I'm here to help with your travel plans.

You: what is my name?
TravelBuddy: You told me your name is bibi!

You: I am feeling great
TravelBuddy: That's great to hear! Are you planning to go on a trip soon?

You: I want to go for a holiday
TravelBuddy: Which continent are you thinking of going to?

You: I want to go to Africa
TravelBuddy: Africa has these countries: Egypt, Kenya, Morocco. Which one interests you?

You: I want to go to Kenya
TravelBuddy: Kenya is in Africa.
Cities: Nairobi, Mombasa, Kisumu
Things to do: Maasai Mara safari, Diani Beach, Nairobi National Park
Visa info: UK: eVisa required
Difficulty: Moderate – eVisa needed
Budget: low: £30-50/day, medium: £60-100/day, high: £120+/day
Flights: Kenya Airways, Emirates, Qatar Airways

You: What's the weather in Kenya?
TravelBuddy: I can't fetch live weather, but I hope it's sunny in Kenya!

You: How much does it cost to go to Kenya?
TravelBuddy: Costs for Kenya vary, but you can save money by eating local food, using public transport, and staying in hostels or guesthouses.

You: What flights go to Kenya?
TravelBuddy: For flights to Kenya, try checking Skyscanner or Google Flights. They usually show the best budget and fastest options.

You: quit
TravelBuddy: Goodbye! Safe travels!
```

(Screenshot of the transcript)

4. Pattern Matching Implementation

Named Regex Groups

The chatbot will use name groups to extract the variables from the user input:

"i am (?P<feeling>doing great|amazing|good|fantastic)"

Element and Segment matching

Used to identify countries:

"i want to go to (?P<country>[A-Za-z]+)"

Multi pattern intent

Each intent has multiple patterns so the TravelBuddy can recognise the variations

Fallback Pattern

When no patterns match, a fallback is used to keep the conversation flowing

5. Advanced Features

Memory System:

The chatbot will remember the user's name and will be able to recall it later

It stores the name in a dictionary when its recognised and the puts the name into future responses

Text Processing:

TravelBuddy will take the country/ continent names from the user, if its in the travel data, it will go there and if it isn't, it will generate a generic but helpful travel response

Regex Bases NLP Parsing:

Named groups, Multi variable extractions and Flexible phrase recognition

Automated Travel info:

For the countries in database, the bot will automatically generate the information such as cities, things to do, visa rules, budget and flights

6. Timeline:

Week 3: Having a Basic understanding of pattern matching through regex and dictionary, had the basic structure and tested with simple greetings and questions

Week 4: Using JSON file, with intents and travel data,integrated pattern matching with responses

Week 5: Added memory system, with country and continent info, fallback handling, refined structured